

Temporal Hallucination Experiment

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Model: Qwen3-1.7B via Ollama | Dataset: TempLAMA (2020 held-out)

1. Project Status Summary

This checkpoint documents the Halulujah project's investigation into whether 'hallucination' settings (higher temperature, top_p, top_k) can help LLMs discover temporal facts that changed between years - facts that would be 'unknown' if the model was only trained on prior years' data.

Core hypothesis: Higher-temperature sampling might enable 'creative' random guessing that finds correct answers at a higher rate for changed facts vs. same facts.

2. Key Experiments Completed

A. TempLAMA Ablation Study (n=50, qwen3:1.7b)

- * bare (no context): 4.0% accuracy (2/50)
- * context (entity history): 70.0% accuracy (35/50)
- * cot (chain-of-thought, no context): 6.0% accuracy (3/50)
- * cot+context: 70.0% accuracy (35/50)

Finding: Context is the dominant factor. CoT alone doesn't help.

B. Hallucination Sweep (128 combinations, ~10 hours)

Grid: 8 temps (0.1-2.0) x 4 top_p (0.5-1.0) x 2 top_k (20,50) x 2 modes

Stratified sample: 10 changed facts + 88 same facts from 2020

Results by temperature (averaged):

Temp	Changed-Acc
0.1	27.9%
0.3	26.8%
0.5	24.8%
0.7	24.8%
1.0	24.3%
1.3	27.1%
1.6	25.0%
2.0	24.9%

CONCLUSION: Changed-fact accuracy is FLAT across temperatures (25-28%). Higher temperature does NOT produce 'creative' correct guesses. The hallucination-as-creativity hypothesis is NOT supported.

3. SSRL Paper Analysis & Alignment

The SSRL (Self-Search Reinforcement Learning) paper suggests our experimental design may be missing key mechanisms:

- * Pass@k scaling: SSRL shows pass@k performance increases significantly with repeated sampling. We only tested single-shot generation.

- * Information masking: SSRL masks key tokens during training to force reliance on learned world knowledge. Our context provides answers explicitly.

- * Format rewards: SSRL uses format-based + rule-based rewards. Our verification uses fuzzy string matching.

- * Sim-to-real: SSRL trains on simulated QA then tests on real benchmarks. We're testing raw inference without RL.

KEY GAP: We tested temperature variation for single-shot. SSRL suggests testing pass@k (repeated sampling) to see if correct answers appear in ANY of k samples.

4. Recommended Next Steps

Based on SSRL insights and time-series causal reasoning literature:

1. Pass@k sampling: Generate k=16,32,64 responses per question, check if ANY is correct (pass@k) vs. majority voting (maj@k).

2. Information masking: Train with entity names masked in context to force pattern inference rather than copying.

3. Time-R1 approach: Two-stage training with CoT-based SFT followed by RL on temporal reasoning traces.

4. Self-consistency decoding: Sample multiple reasoning paths at moderate temperature and aggregate.

5. Technical Implementation Notes

Key files:

- * src/scripts/run_templama_sweep.py - Ollama sweep with per-question timeouts
- * src/scripts/run_templama_ablation.py - 4-condition ablation study
- * src/halulujah/data/loader.py - TempLAMA loader with entity history context
- * src/halulujah/rl/verifier.py - Multi-answer verification

Outputs:

- * outputs/templama_ablation_2020_qwen3_1.7b/ - Ablation results
- * outputs/templama_sweep_2020_qwen3_1.7b/ - Full 128-combo sweep results

6. Where We Left Off

CURRENT STATE: The hallucination-as-creativity hypothesis is NOT supported. Temperature sweeps (0.1 to 2.0) show flat changed-fact accuracy around 25-28%.

The SSRL paper suggests a pivot: instead of hoping temperature variation helps, we should:

1. Test pass@k sampling (does the correct answer appear in ANY of k samples?)
2. Consider RL training with format/rule rewards
3. Explore information masking during fine-tuning

The next session should implement a pass@k experiment on the same TempLAMA sample to determine if repeated sampling surfaces correct answers for changed facts that single-shot generation misses.

7. References

- * SSRL: Self-Search Reinforcement Learning (references/)
- * TempLAMA: Temporal Knowledge Probing (Dhingra et al., 2022)
- * Time-R1: Two-stage temporal reasoning (2025)
- * CAIFormer: Causal-Aware Transformer for Time Series (KDD 2025)